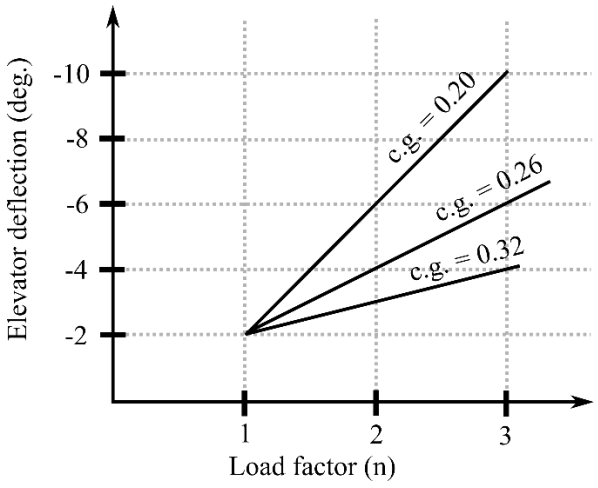
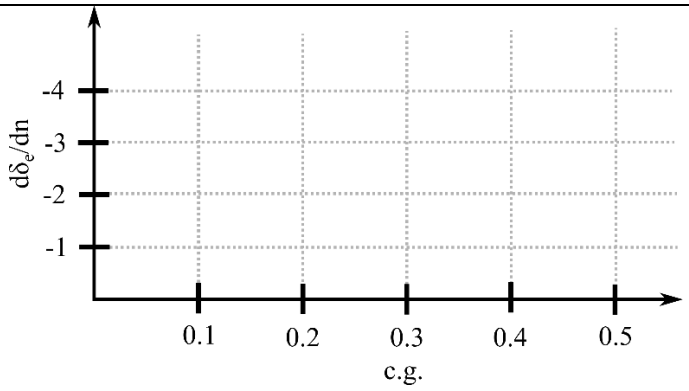
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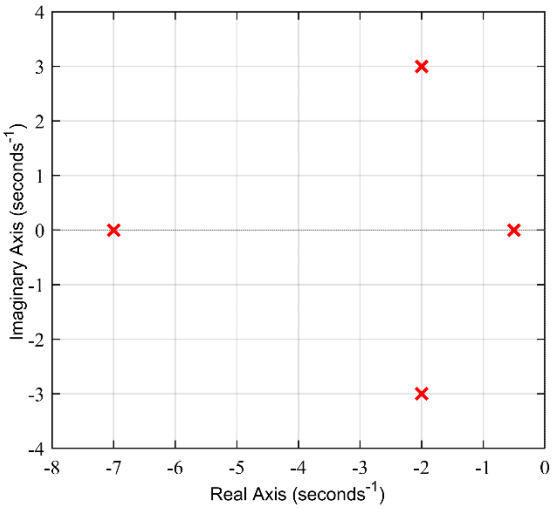
DEPARTMENT OF AERONAUTICAL AND AUTOMOBILE ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Aeronautical Engineering**	**Course Type**	**C**
Course Name	**Flight Dynamics and Control**	**Course Code**	**AAE 3121**
Semester	**V**	**Max Marks**	**50**
Date	**21-11-2024**	**Duration**	**180 mins**

Instructions:
- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET		BL
			CO	PO	CLO	AHEP4 LO	
1A	Consider the following aircraft flying in straight and level unaccelerated trimmed flight at M = 0.2 at sea-level. $x_{cg} = 0.5\bar{c}_{wing};\ M_{ac_{tail}} = 0;\ M_{ac_{wing}} = -6779$ Nm; T $= 13345$ N W $= 53379$ N; $\bar{c}_{wing} = 2.45$ m; $\bar{c}_{tail} = 1.22$ m; $b_{wing} = 9.14$ m Distance from the c.g. to the leading edge of the horizontal tail is given as 3.66 m. Evaluate the aircraft's aerodynamic characteristics by addressing the following tasks: a. Determine the lift generated by both the wing and the tail, given that they act at their respective quarter chord points. b. Analyze and compute the drag force acting on the aircraft. c. Using the assumption of a rectangular wing, calculate the overall lift coefficient for the entire aircraft. d. Calculate the pitching moment coefficient about the center of gravity for the complete aircraft configuration.	**(05)**	02	1, 2, 4	**02**	**C1, C2, C3**	**05**

1B	In what ways does the concept of trim differ from equilibrium in the contexts of steady turns, steady rolls, and steady climbs?	**(03)**	02	1, 2, 4	**02**	**C1, C2, C3**	**02**
1C	Explain the static stability criterion for rolling, pitching, and yawing motions?	**(02)**	02	1, 2, 4	02	C1, C2, C3	**02**
2A	Design a solution to determine the necessary horizontal tail area for an aircraft that meets a specified static margin of $0.08\bar{c}$, given the following conditions: The aerodynamic center (a.c.) of the wing is located at $0.2\ \bar{c}$ and the c.g. of the aircraft is at $0.25\ \bar{c}$. Wing lift curve slope $(C_{L_{\alpha,w}})$ = 0.1/deg; Tail lift curve slope $(C_{L_{\alpha,t}})$ = 0.08/deg; Downwash angle at the tail (ϵ) = 0.45α; Tail moment arm (l_t) = $2.5\ \bar{c}$; Tail efficiency factor (η_T) = 0.95; Wing area (S_W) = 25 m^2. Integrate these parameters to develop a calculated horizontal tail area that ensures the aircraft's stability and control requirements are met.	**(05)**	02	1, 2, 4	**02**	**C1, C2, C3**	**05**
2B	Analyze the provided plot to determine the change in elevator deflection per unit load factor, $(d\delta_e/dn)$ for each center of gravity (c.g.) location. Using this data, plot the values on the blank graph (where the x-axis represents the c.g. location in terms of the mean aerodynamic chord of the wing) and interpret the resulting graph to estimate the maneuver point.  	**(03)**	02	1, 2, 4	**02**	**C1, C2, C3**	**03**

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<tr><td>2C</td><td>Explain two design approaches that can be used to enhance roll static stability.</td><td>(02)</td><td>02</td><td>1, 2, 4</td><td>02</td><td>C1, C2, C3</td><td>02</td></tr>
<tr><td>3A</td><td>Evaluate how the pure lift force of 50 N, given after a 10-degree pitch-up and a 20-degree yaw, can be decomposed into its force vector components within the inertial reference frame after transformation from the body frame.</td><td>(05)</td><td>03</td><td>1, 2, 4</td><td>03</td><td>C1, C2, C3</td><td>05</td></tr>
<tr><td>3B</td><td>A twin-engine jet aircraft has a thrust of 20 kN per engine and the engines are separated by a distance of 10 m. The wing area is 60 m^2 and wing span 15 m. Assuming maximum rudder deflection available is $\pm 25°$, determine minimum rudder effectiveness $C_{N_{\delta r}}$ required to hold the aircraft at zero sideslip with right engine failed at a flight speed of 75 m/s at sea-level. State whether $C_{N_{\delta r}}$ is positive or negative and sense of rudder deflection required.</td><td>(03)</td><td>02</td><td>1, 2, 4</td><td>02</td><td>C1, C2, C3</td><td>04</td></tr>
<tr><td>3C</td><td>Explain how the Mach tuck condition occurs and analyze the design strategies used in the Concorde supersonic transport aircraft to address this aerodynamic challenge.</td><td>(02)</td><td>02</td><td>1, 2, 4</td><td>02</td><td>C1, C2, C3</td><td>03</td></tr>
<tr><td>4A</td><td>Identify the effects on the aircraft's orientation or control surfaces that correspond to each of the following aircraft movements or control inputs:
<table>
<tr><td>(1)</td><td>Positive pitching</td><td>(a)</td><td>Right wing down</td></tr>
<tr><td>(2)</td><td>Positive Y-axis</td><td>(b)</td><td>Right wing retreating</td></tr>
<tr><td>(3)</td><td>Positive rolling</td><td>(c)</td><td>Elevator down</td></tr>
<tr><td>(4)</td><td>Positive yawing</td><td>(d)</td><td>Left wing down</td></tr>
<tr><td>(5)</td><td>Negative rolling</td><td>(e)</td><td>Towards right wing</td></tr>
<tr><td>(6)</td><td>Negative yawing</td><td>(f)</td><td>Negative yawing</td></tr>
<tr><td>(7)</td><td>Positive sideslip</td><td>(g)</td><td>Elevator up</td></tr>
</table></td><td>(04)</td><td>01</td><td>1, 2, 4</td><td>01</td><td>C1, C2, C3</td><td>03</td></tr>
</table>

	(8) Negative pitching (h) Right wing advancing							
4B	An airplane flying at 4572 m, Mach 0.6 with an alpha trim of 3.4 degree has the following short period characteristics: short period damping ratio = 0.304 and short period natural frequency = 3.69 rad/s. Given that the pilot has made a unit step elevator input, analyze the response of the airplane's angle of attack in terms of a second-order system by determining: a. The time required for the airplane to reach its first overshoot, b. The magnitude of the first overshoot, and c. The duration required for the angle of attack to fall within $\pm$2% of its final steady-state value.	**(03)**	04	1, 2, 4	**04**	**C1, C2, C3**	**05**
4C	Figure shows the stick fixed lateral roots of a jet transport airplane. Identify the roots and determine the time for the amplitude and period to halve or double where applicable.  	**(03)**	04	1, 2, 4	**04**	**C1, C2, C3**	**04**
5A	Consider the following longitudinal dynamics of an aircraft at some trim condition with airspeed of 502 ft/s, where $\Delta x = [\Delta u, \Delta w, \Delta q, \Delta\Theta]^T$: $\Delta\dot{x} = \begin{bmatrix} \Delta\dot{u} \\ \Delta\dot{w} \\ \Delta\dot{q} \\ \Delta\dot{\Theta} \end{bmatrix} = \begin{bmatrix} -0.020 & 0.016 & -0.65 & -32.17 \\ -0.13 & -1.019 & 454.21 & 0 \\ 0 & -0.0050 & -1.38 & 0 \\ 0 & 0 & 1.0 & 0 \end{bmatrix} \begin{bmatrix} \Delta u \\ \Delta w \\ \Delta q \\ \Delta\Theta \end{bmatrix}$ Determine the characteristic equation by synthesizing the information provided from the given stability derivatives.	**(04)**	03	1, 2, 4	**03**	**C1, C2, C3**	**05**
5B	Determine the eigenvalues associated with the short period and phugoid modes by analyzing the characteristic equation derived in question 5(A) using Lin's method.	**(03)**	03	1, 2, 4	**03**	**C1, C2, C3**	**04**

5C	Analyze the roots obtained in 5(B) to deduce the natural frequency and damping ratio for both the short period mode and the phugoid mode.	**(03)**	03	1, 2, 4	**03**	**C1, C2, C3**	**05**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***04***
2	***25***
3	***15***
4	***06***
5	**---**

BL	Total Marks
2	**07**
3	**09**
4	**09**
5	**25**